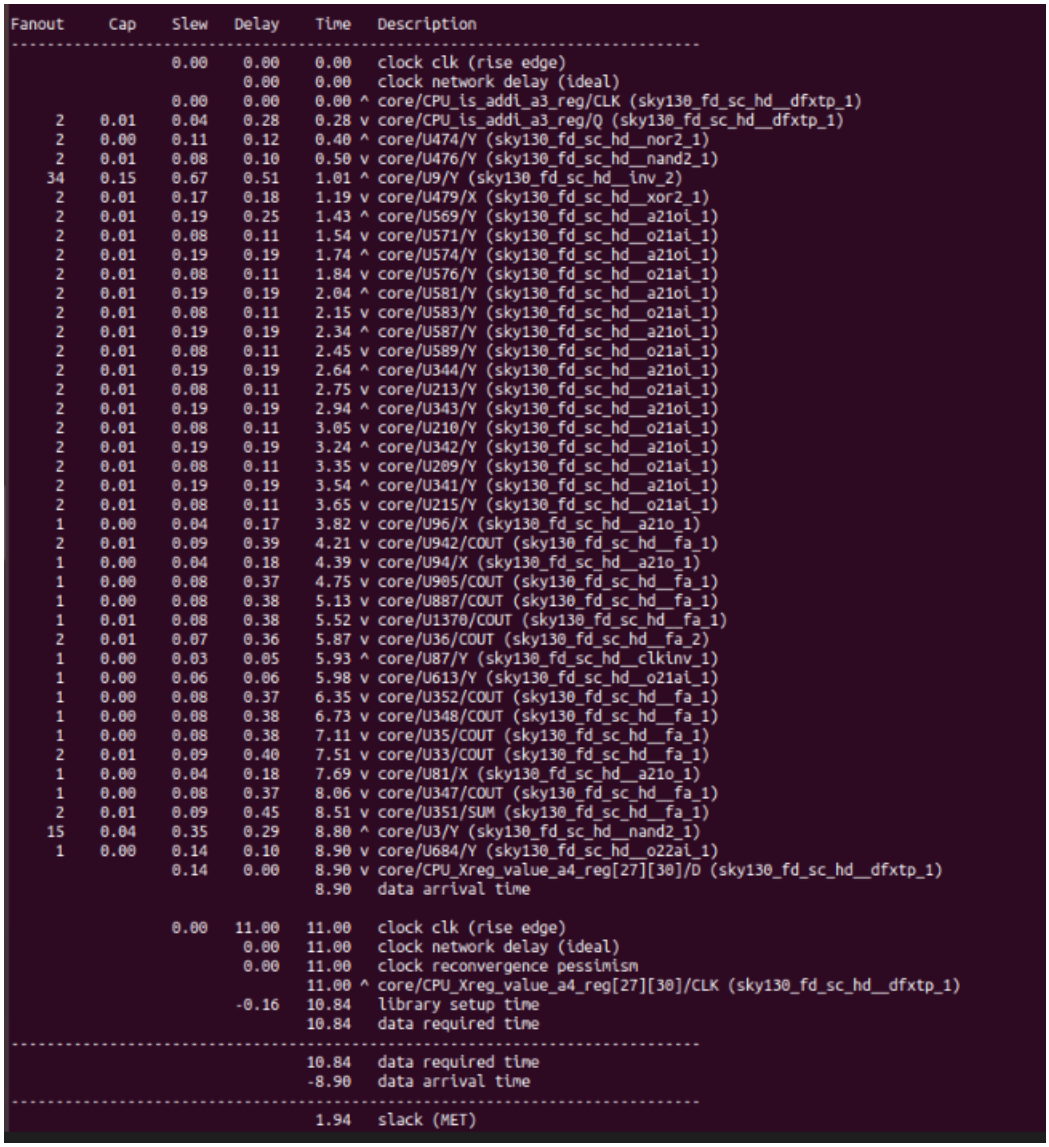
OPENSTA LIBRARY OUTPUT ANALYSIS
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Note: These are analyses of the exact terminal results supplied. All reported paths are Path Type: max, so the slack discussed below is setup/max-delay slack.
The terminal output also contains 'library already exists' warnings because the Tcl loop loads libraries repeatedly in the same OpenSTA session. Those warnings do not themselves indicate timing violations, but they mean the runs are not as cleanly isolated as separate OpenSTA processes. The numerical results below are the results actually printed by the supplied run.

1. sky130_fd_sc_hd__tt_025C_1v80.lib

The library name indicates the SkyWater 130 nm process (`sky130`), the high-density standard-cell library (`fd_sc_hd`), the Typical-Typical process corner (`tt`), a temperature of 25°C (`025C`), and a supply voltage of 1.80 V (`1v80`). This is the nominal/typical operating condition and provides a useful reference for the other corners. OpenSTA reports a maximum-delay (setup) path from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`, both clocked by `clk`. The data arrival time is 8.90 ns, while the data required time is 10.84 ns, giving a slack of +1.94 ns. Since the slack is positive and the report says `slack (MET)`, there is no setup violation for this reported path. The path has 1.94 ns of timing margin before it would fail the 11 ns clock-period constraint. The path contains a long combinational chain, including several AOI/OAI cells, full adders, and logic gates, which accounts for most of the accumulated delay. The result therefore shows that under the typical 25°C, 1.80 V condition, this reported critical path completes comfortably within the available clock period.



```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.11    0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.08    0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.15    0.67    0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.17    0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.19    0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.08    0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.04    0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.09    0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.01    0.08    0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.07    0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.03    0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.06    0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.08    0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.04    0.35    0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.14    0.10    8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.14    0.00    8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.90   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                         0.00   11.00   clock network delay (ideal)
                         0.00   11.00   clock reconvergence pessimism
                                11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                        -0.16   10.84   library setup time
                                10.84   data required time
---------------------------------------------------------------------------
                                10.84   data required time
                                -8.90   data arrival time
---------------------------------------------------------------------------
                                 1.94   slack (MET)
```

2. sky130_fd_sc_hd__tt_100C_1v80.lib

The library name identifies the Sky130 high-density standard-cell library at the Typical-Typical (`tt`) process corner, 100°C (`100C`), and 1.80 V (`1v80`). Compared with the previous TT corner, only the temperature changes from 25°C to 100°C. The reported maximum-delay path again starts at `core/CPU_is_addi_a3_reg` and ends at `core/CPU_Xreg_value_a4_reg[27][30]`. Its data arrival time is 8.81 ns and the data required time is 10.90 ns, producing a slack of +2.09 ns. The path therefore meets the setup requirement with 2.09 ns of margin and has no reported violation. Interestingly, the arrival time is 0.09 ns smaller than in the 25°C TT result, so this particular characterized library does not show the simple assumption that increasing temperature must always increase the reported path delay. The required time also changes because the library setup time changes from 0.16 ns to 0.10 ns. The important point is that the actual OpenSTA characterization gives a positive slack, so the reported path satisfies the timing constraint at this corner.

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.29    0.29 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.11    0.11    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.09    0.10    0.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.15    0.64    0.48    0.99 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.18    0.19    1.18 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.19    0.23    1.41 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.12    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    2.31 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    2.42 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    2.61 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    2.72 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    2.90 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    3.02 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    3.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    3.32 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.19    0.18    3.50 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.11    3.62 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.05    0.17    3.79 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.10    0.38    4.16 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.05    0.17    4.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.36    4.69 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.01    0.09    0.37    5.43 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.08    0.35    5.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.03    0.05    5.83 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.07    0.06    5.89 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.08    0.36    6.26 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    6.62 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    6.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.10    0.39    7.38 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.05    0.17    7.55 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.36    7.91 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.08    0.49    8.40 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    0.34    0.25    8.65 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.20    0.16    8.81 ^ core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.20    0.00    8.81 ^ core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.81   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                         0.00   11.00   clock network delay (ideal)
                         0.00   11.00   clock reconvergence pessimism
                                11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                        -0.10   10.90   library setup time
                                10.90   data required time
---------------------------------------------------------------------------
                                10.90   data required time
                                -8.81   data arrival time
---------------------------------------------------------------------------
                                 2.09   slack (MET)
```

3. sky130_fd_sc_hd__ss_n40C_1v28.lib

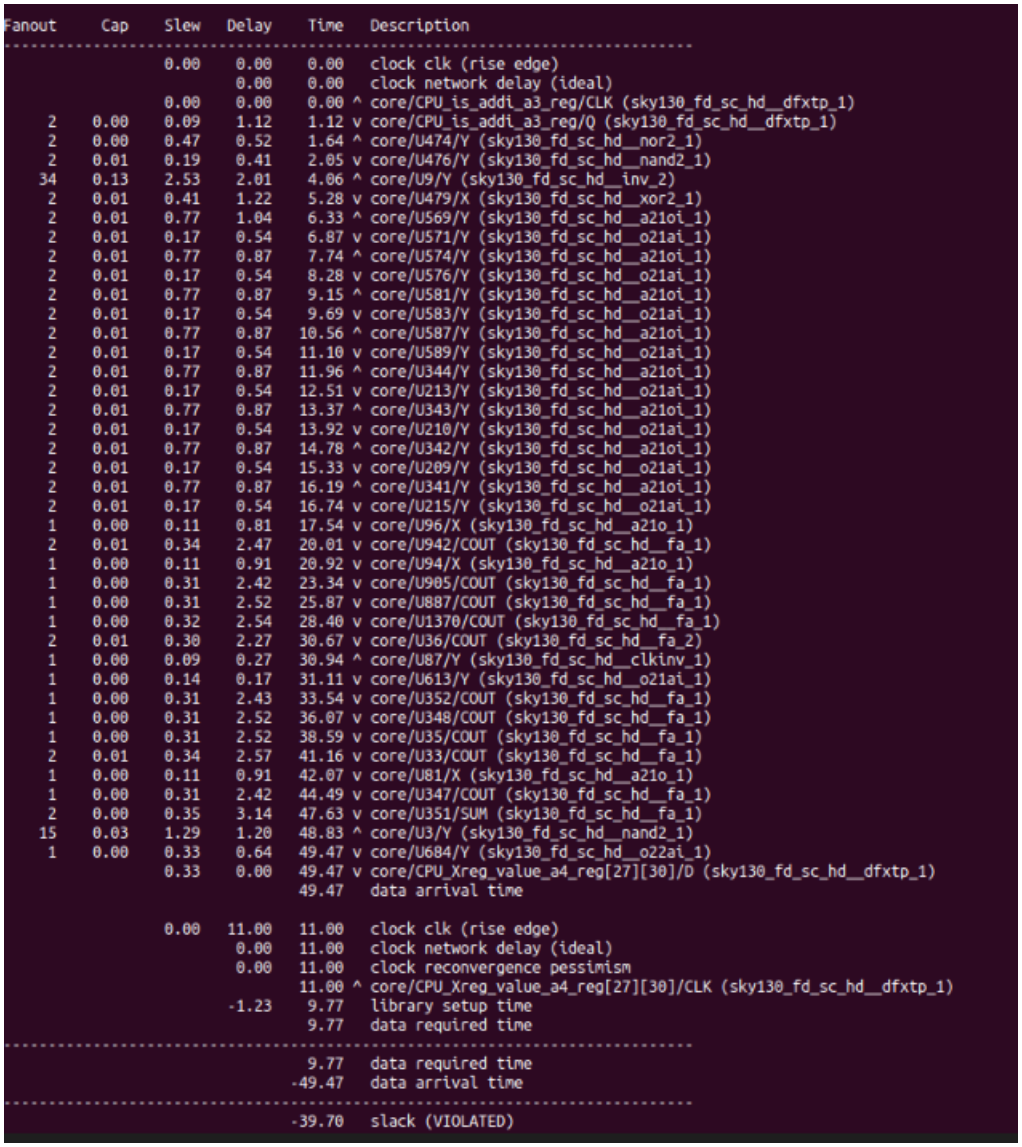
The library name denotes the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.28 V (`1v28`). This is a slow-process, low-voltage operating condition. The maximum-delay path is from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 71.70 ns, while the required time is only 9.14 ns. This produces a very large negative slack of −62.56 ns, and OpenSTA reports `slack (VIOLATED)`. Therefore, the path misses the required arrival time by 62.56 ns, which is a severe setup-timing violation. The very large delay is visible throughout the path: many individual cells have substantially larger delays and slews than in the TT results, and the accumulated combinational delay reaches 71.70 ns. The low 1.28 V supply significantly reduces the available drive strength in this characterized corner, while the SS process also represents slow device behavior. With the clock period fixed at 11 ns, the design cannot satisfy the setup requirement for this reported path under this PVT condition.

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.11    1.50    1.50 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.65    0.71    2.20 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.23    0.56    2.76 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.13    3.38    2.67    5.43 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.53    1.71    7.14 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    1.07    1.42    8.57 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76    9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   14.45 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   15.21 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   16.41 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   17.17 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   18.38 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   19.14 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   20.34 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   21.10 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    1.07    1.20   22.30 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.22    0.76   23.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.15    1.13   24.19 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.49    3.70   27.89 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.15    1.32   29.21 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.45    3.62   32.83 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.45    3.78   36.61 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.46    3.80   40.42 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.44    3.39   43.81 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.12    0.40   44.21 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.17    0.21   44.43 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.45    3.63   48.06 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.45    3.78   51.84 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.45    3.78   55.62 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.49    3.86   59.48 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.15    1.32   60.80 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.45    3.62   64.42 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.51    4.76   69.19 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    1.70    1.61   70.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.42    0.91   71.70 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.42    0.00   71.70 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          71.70   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -1.86    9.14   library setup time
                                  9.14   data required time
---------------------------------------------------------------------------
                                  9.14   data required time
                                -71.70   data arrival time
---------------------------------------------------------------------------
                                -62.56   slack (VIOLATED)
```

4. sky130_fd_sc_hd__ss_n40C_1v35.lib

The library name represents the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.35 V (`1v35`). It is therefore another low-voltage slow-process condition, with slightly more supply voltage than the 1.28 V corner. The reported maximum-delay path is again from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 49.47 ns and the data required time is 9.77 ns, resulting in a slack of −39.70 ns. OpenSTA marks the path as `VIOLATED`, so the setup requirement is missed by 39.70 ns. Compared with the 1.28 V SS result, increasing the supply from 1.28 V to 1.35 V substantially reduces the path delay, from 71.70 ns to 49.47 ns, and improves slack by 22.86 ns. However, the slack remains strongly negative. This shows the strong effect of supply voltage on the delay of this long combinational path under the SS process corner.



```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.09    1.12    1.12 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.47    0.52    1.64 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.19    0.41    2.05 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.13    2.53    2.01    4.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.41    1.22    5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.77    1.04    6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54    6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87    7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54    8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87    9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54    9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87   10.56 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54   11.10 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87   11.96 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54   12.51 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87   13.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54   13.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87   14.78 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54   15.33 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.77    0.87   16.19 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.17    0.54   16.74 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.11    0.81   17.54 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.34    2.47   20.01 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.91   20.92 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.31    2.42   23.34 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.31    2.52   25.87 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.32    2.54   28.40 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.30    2.27   30.67 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.09    0.27   30.94 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.14    0.17   31.11 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.31    2.43   33.54 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.31    2.52   36.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.31    2.52   38.59 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.34    2.57   41.16 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.91   42.07 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.31    2.42   44.49 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.35    3.14   47.63 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    1.29    1.20   48.83 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.33    0.64   49.47 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.33    0.00   49.47 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          49.47   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -1.23    9.77   library setup time
                                  9.77   data required time
---------------------------------------------------------------------------
                                  9.77   data required time
                                -49.47   data arrival time
---------------------------------------------------------------------------
                                -39.70   slack (VIOLATED)
```

5. sky130_fd_sc_hd__ss_n40C_1v40.lib

The library name indicates the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.40 V (`1v40`). The reported maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 39.74 ns and the data required time is 10.04 ns, giving a slack of −29.69 ns. Because the slack is negative, OpenSTA reports `slack (VIOLATED)`, meaning the path misses the setup requirement by 29.69 ns. Compared with the 1.35 V SS corner, the higher 1.40 V supply reduces the arrival time from 49.47 ns to 39.74 ns and improves the slack by about 10.01 ns. Nevertheless, the path remains far outside the 11 ns clock requirement. The repeated chain of logic cells and arithmetic carry stages is responsible for the large accumulated delay, while the SS process continues to represent slow cell behavior.

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.08    0.94    0.94 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.39    0.43    1.38 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.17    0.34    1.72 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.14    2.13    1.70    3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.36    0.99    4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.64    0.87    5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44    5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71    6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44    6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71    7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44    8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71    8.74 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44    9.18 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71    9.89 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44   10.33 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71   11.05 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44   11.49 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71   12.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44   12.64 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.64    0.71   13.36 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.15    0.44   13.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.10    0.66   14.46 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.28    1.95   16.41 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.10    0.74   17.14 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.25    1.91   19.05 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.25    1.98   21.04 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.26    2.00   23.03 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.24    1.79   24.82 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.08    0.22   25.04 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.12    0.14   25.18 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.25    1.92   27.10 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.25    1.98   29.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.25    1.98   31.07 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.28    2.02   33.10 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.10    0.74   33.83 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.25    1.91   35.74 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.28    2.46   38.20 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    1.09    1.00   39.20 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.29    0.53   39.74 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.29    0.00   39.74 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          39.74   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.96   10.04   library setup time
                                 10.04   data required time
---------------------------------------------------------------------------
                                 10.04   data required time
                                -39.74   data arrival time
---------------------------------------------------------------------------
                                -29.69   slack (VIOLATED)
```

6. sky130_fd_sc_hd__ss_n40C_1v44.lib

The library name identifies the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.44 V (`1v44`). The maximum-delay path again runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 34.10 ns and the data required time is 10.20 ns, resulting in a slack of −23.91 ns. The report therefore indicates a setup violation of 23.91 ns. Increasing the voltage from 1.40 V to 1.44 V reduces the reported arrival time by about 5.64 ns and improves the slack by 5.78 ns, showing a clear voltage-dependent improvement in this SS group. However, the path still takes much longer than the approximately 10 ns available data time after setup requirements are considered. The long combinational chain, together with the slow SS characterization, dominates the timing result.

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.07    0.83    0.83 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.34    0.38    1.21 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.15    0.30    1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.14    1.89    1.50    3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.34    0.88    3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.56    0.77    4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39    5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62    5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39    6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62    6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39    7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62    7.69 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39    8.08 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62    8.71 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39    9.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62    9.72 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39   10.11 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62   10.73 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39   11.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.56    0.62   11.75 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.39   12.14 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.09    0.57   12.71 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.24    1.65   14.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.09    0.63   15.00 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.22    1.61   16.61 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.22    1.67   18.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.22    1.68   19.96 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.20    1.51   21.48 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.07    0.19   21.67 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.11    0.13   21.79 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.22    1.62   23.42 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.22    1.67   25.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.22    1.67   26.76 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.24    1.71   28.47 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.09    0.63   29.10 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.22    1.61   30.72 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.24    2.07   32.79 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    0.96    0.87   33.65 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.26    0.45   34.10 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.26    0.00   34.10 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          34.10   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.80   10.20   library setup time
                                 10.20   data required time
---------------------------------------------------------------------------
                                 10.20   data required time
                                -34.10   data arrival time
---------------------------------------------------------------------------
                                -23.91   slack (VIOLATED)
```

7. sky130_fd_sc_hd__ss_n40C_1v60.lib

The library name denotes the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.60 V (`1v60`). The maximum-delay path is from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 21.15 ns and the required time is 10.54 ns, giving a slack of −10.61 ns. OpenSTA therefore reports a setup violation of 10.61 ns. Compared with the lower-voltage SS corners, the higher 1.60 V supply greatly improves the result: the arrival time is much smaller than the 34.10 ns at 1.44 V, and the negative slack is correspondingly reduced. Even so, the data arrives about 10.61 ns later than the permitted limit. The result demonstrates that raising the supply voltage improves the timing of the slow-process cells, but at this corner the improvement is still insufficient to meet the 11 ns clock constraint.

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.05    0.57    0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.23    0.25    0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.12    0.20    1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.14    1.32    1.04    2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.25    0.53    2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.38    0.52    3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.24    7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.38    0.41    7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.11    0.25    7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.06    0.37    8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.16    0.98    9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.06    0.40    9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.14    0.96   10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.14    0.99   11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.15    1.00   12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.13    0.91   13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.05    0.12   13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.08    0.09   13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.14    0.96   14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.14    0.99   15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.14    0.99   16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.16    1.02   17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.06    0.40   18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.14    0.96   19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.16    1.21   20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.04    0.68    0.59   20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.20    0.29   21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.20    0.00   21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          21.15   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.46   10.54   library setup time
                                 10.54   data required time
---------------------------------------------------------------------------
                                 10.54   data required time
                                -21.15   data arrival time
---------------------------------------------------------------------------
                                -10.61   slack (VIOLATED)
```

8. sky130_fd_sc_hd__ss_n40C_1v76.lib

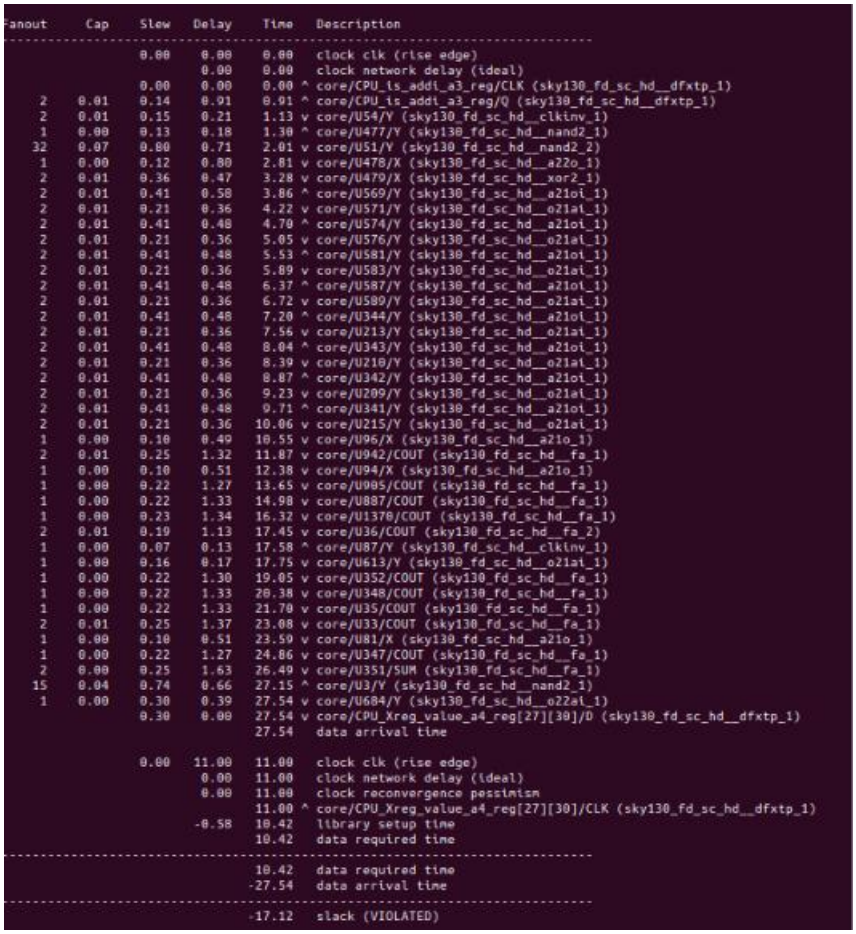
The library name identifies the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, −40°C (`n40C`), and 1.76 V (`1v76`). This is the highest-voltage −40°C SS corner in the supplied set. The reported maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. Its data arrival time is 15.07 ns and the data required time is 10.69 ns, producing a slack of −4.38 ns. The path therefore still has a setup violation, missing the required timing by 4.38 ns. Compared with the 1.60 V SS result, increasing the voltage to 1.76 V reduces the arrival time by 6.08 ns and improves slack by 6.23 ns. This is consistent with the overall voltage trend observed across the SS −40°C corners: higher supply voltage substantially reduces the cell delays. However, even the 1.76 V SS condition does not meet the required timing for this particular long path.

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.05    0.43    0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.17    0.19    0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.10    0.15    0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.15    1.02    0.80    1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.21    0.36    1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.29    0.39    2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    3.77 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    3.95 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    4.26 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    4.44 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    4.74 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    4.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    5.23 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    5.41 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.29    0.31    5.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.09    0.18    5.89 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.05    0.27    6.16 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.12    0.68    6.84 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.05    0.29    7.13 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.11    0.65    7.79 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.68    8.47 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.68    9.15 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.10    0.63    9.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.04    0.09    9.87 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.07    0.07    9.94 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.11    0.66   10.61 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.68   11.28 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.11    0.68   11.96 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.12    0.70   12.66 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.05    0.29   12.95 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.11    0.65   13.61 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.00    0.12    0.82   14.43 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.04    0.52    0.45   14.88 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.17    0.19   15.07 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.17    0.00   15.07 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          15.07   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.31   10.69   library setup time
                                 10.69   data required time
---------------------------------------------------------------------------
                                 10.69   data required time
                                -15.07   data arrival time
---------------------------------------------------------------------------
                                 -4.38   slack (VIOLATED)
```

9. sky130_fd_sc_hd__ss_100C_1v40.lib

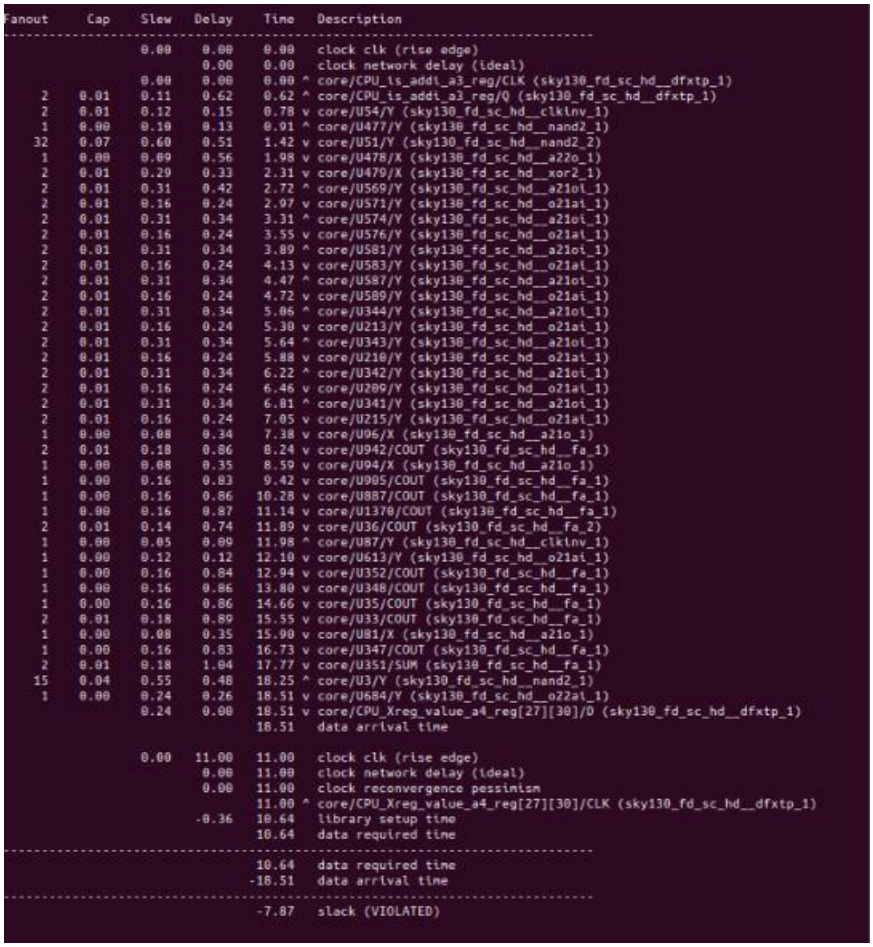
The library name represents the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, 100°C (`100C`), and 1.40 V (`1v40`). The reported maximum-delay path is from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 27.54 ns, while the data required time is 10.42 ns. The resulting slack is −17.12 ns, so OpenSTA reports a setup violation of 17.12 ns. Compared with the −40°C, 1.40 V SS result, the 100°C corner has a smaller reported arrival time in this particular characterization, but it still fails the same clock constraint. The required time is also slightly different because the endpoint setup time changes with the library corner. The important observation from the actual output is that the SS process at 1.40 V produces a substantially delayed path and negative timing margin, regardless of the temperature difference. The long combinational logic chain remains the dominant source of the path delay.



Fanout	Cap	Slew	Delay	Time	Description
		0.00	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
		0.00	0.00	0.00 ^	core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.14	0.91	0.91 ^	core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.15	0.21	1.13 v	core/U54/Y (sky130_fd_sc_hd__clkinv_1)
1	0.00	0.13	0.18	1.30 ^	core/U477/Y (sky130_fd_sc_hd__nand2_1)
32	0.07	0.80	0.71	2.01 v	core/U51/Y (sky130_fd_sc_hd__nand2_2)
1	0.00	0.12	0.80	2.81 v	core/U478/X (sky130_fd_sc_hd__a22o_1)
2	0.01	0.36	0.47	3.28 v	core/U479/X (sky130_fd_sc_hd__xor2_1)
2	0.01	0.41	0.58	3.86 ^	core/U569/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	4.22 v	core/U571/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	4.70 ^	core/U574/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	5.05 v	core/U576/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	5.53 ^	core/U581/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	5.89 v	core/U583/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	6.37 ^	core/U587/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	6.72 v	core/U589/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	7.20 ^	core/U344/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	7.56 v	core/U213/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	8.04 ^	core/U343/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	8.39 v	core/U210/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	8.87 ^	core/U342/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	9.23 v	core/U209/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.41	0.48	9.71 ^	core/U341/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.21	0.36	10.06 v	core/U215/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00	0.10	0.49	10.55 v	core/U96/X (sky130_fd_sc_hd__a21o_1)
2	0.01	0.25	1.32	11.87 v	core/U942/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.10	0.51	12.38 v	core/U94/X (sky130_fd_sc_hd__a21o_1)
1	0.00	0.22	1.27	13.65 v	core/U905/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.22	1.33	14.98 v	core/U887/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.23	1.34	16.32 v	core/U1370/COUT (sky130_fd_sc_hd__fa_1)
2	0.01	0.19	1.13	17.45 v	core/U36/COUT (sky130_fd_sc_hd__fa_2)
1	0.00	0.07	0.13	17.58 ^	core/U87/Y (sky130_fd_sc_hd__clkinv_1)
1	0.00	0.16	0.17	17.75 v	core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00	0.22	1.30	19.05 v	core/U352/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.22	1.33	20.38 v	core/U348/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.22	1.33	21.70 v	core/U35/COUT (sky130_fd_sc_hd__fa_1)
2	0.01	0.25	1.37	23.08 v	core/U33/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.10	0.51	23.59 v	core/U81/X (sky130_fd_sc_hd__a21o_1)
1	0.00	0.22	1.27	24.86 v	core/U347/COUT (sky130_fd_sc_hd__fa_1)
2	0.00	0.25	1.63	26.49 v	core/U351/SUM (sky130_fd_sc_hd__fa_1)
15	0.04	0.74	0.66	27.15 ^	core/U3/Y (sky130_fd_sc_hd__nand2_1)
1	0.00	0.30	0.39	27.54 v	core/U684/Y (sky130_fd_sc_hd__o22ai_1)
		0.30	0.00	27.54 v	core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
				27.54	data arrival time
		0.00	11.00	11.00	clock clk (rise edge)
			0.00	11.00	clock network delay (ideal)
			0.00	11.00	clock reconvergence pessimism
				11.00 ^	core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
			-0.58	10.42	library setup time
				10.42	data required time
				10.42	data required time
				-27.54	data arrival time
				-17.12	slack (VIOLATED)

10. sky130_fd_sc_hd__ss_100C_1v60.lib

The library name indicates the Sky130 high-density standard-cell library at the Slow-Slow (`ss`) process corner, 100°C (`100C`), and 1.60 V (`1v60`). The maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[27][30]`. The data arrival time is 18.51 ns and the data required time is 10.64 ns, producing a slack of −7.87 ns. This is a setup violation of 7.87 ns because the data arrives later than the allowed time. Raising the voltage from 1.40 V to 1.60 V at the same 100°C SS condition reduces the arrival time from 27.54 ns to 18.51 ns and improves the slack by 9.25 ns. Thus the output clearly shows the timing benefit of the higher supply voltage. Nevertheless, the path still does not meet the 11 ns clock requirement. The result is consistent with the slow-process characterization causing substantially larger delays than the FF corners.

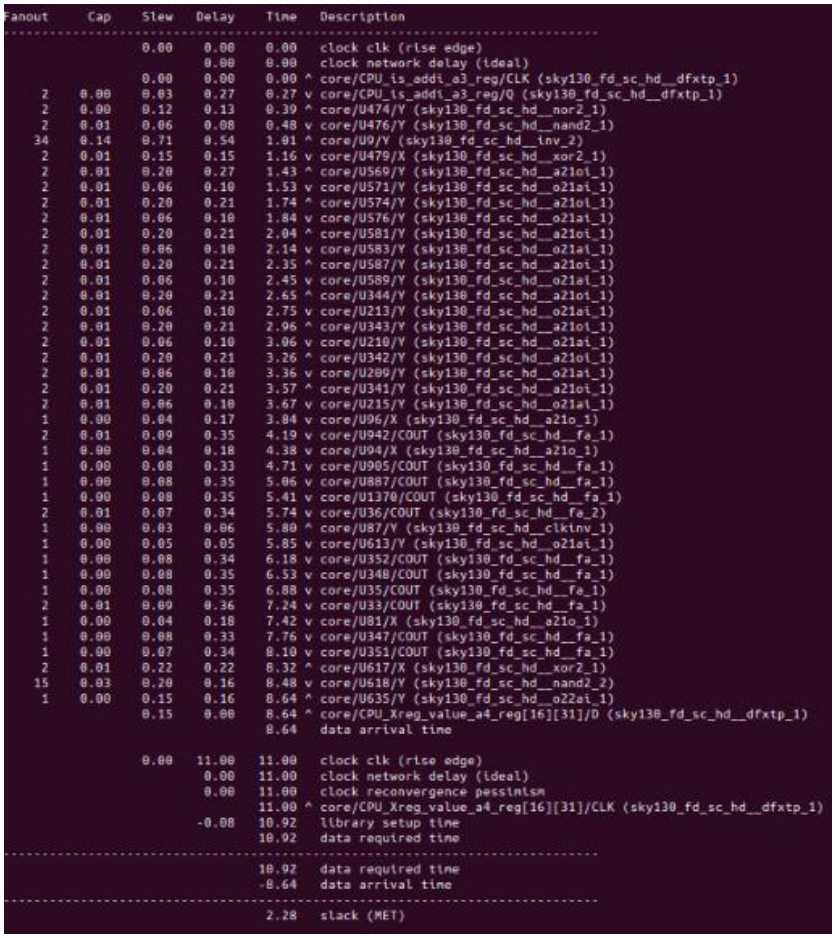


```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.11    0.62    0.62 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.12    0.15    0.78 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.10    0.13    0.91 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
    32    0.07    0.60    0.51    1.42 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
     1    0.00    0.09    0.56    1.98 v core/U478/X (sky130_fd_sc_hd__a22o_1)
     2    0.01    0.29    0.33    2.31 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.31    0.42    2.72 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    2.97 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    3.31 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    3.55 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    4.13 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    4.47 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    4.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    5.06 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    5.30 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    5.64 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    5.88 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    6.22 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    6.46 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.31    0.34    6.81 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.24    7.05 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.08    0.34    7.38 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.18    0.86    8.24 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35    8.59 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.16    0.83    9.42 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.16    0.86   10.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.16    0.87   11.14 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.14    0.74   11.89 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.05    0.09   11.98 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.12    0.12   12.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.16    0.84   12.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.16    0.86   13.80 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.16    0.86   14.66 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.18    0.89   15.55 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35   15.90 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.16    0.83   16.73 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.18    1.04   17.77 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.04    0.55    0.48   18.25 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.24    0.26   18.51 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                  0.24    0.00   18.51 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                                 18.51   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.36   10.64   library setup time
                                 10.64   data required time
-----------------------------------------------------------------------------
                                 10.64   data required time
                                -18.51   data arrival time
-----------------------------------------------------------------------------
                                 -7.87   slack (VIOLATED)
```

11. sky130_fd_sc_hd__ff_n40C_1v56.lib

The library name denotes the Sky130 high-density standard-cell library at the Fast-Fast (`ff`) process corner, −40°C (`n40C`), and 1.56 V (`1v56`). The reported maximum-delay path starts at `core/CPU_is_addi_a3_reg` and ends at `core/CPU_Xreg_value_a4_reg[16][31]`. The data arrival time is 8.64 ns and the required time is 10.92 ns, giving a positive slack of 2.28 ns. OpenSTA therefore reports `slack (MET)` and there is no setup violation for the reported path. The path endpoint differs from the SS/TT reports, showing that the worst reported path can change with the cell timing characteristics. Even at the relatively low 1.56 V supply, the FF process produces much smaller cell delays than the SS corners. The 2.28 ns positive margin means the data reaches the destination 2.28 ns before the setup limit for the 11 ns clock period.

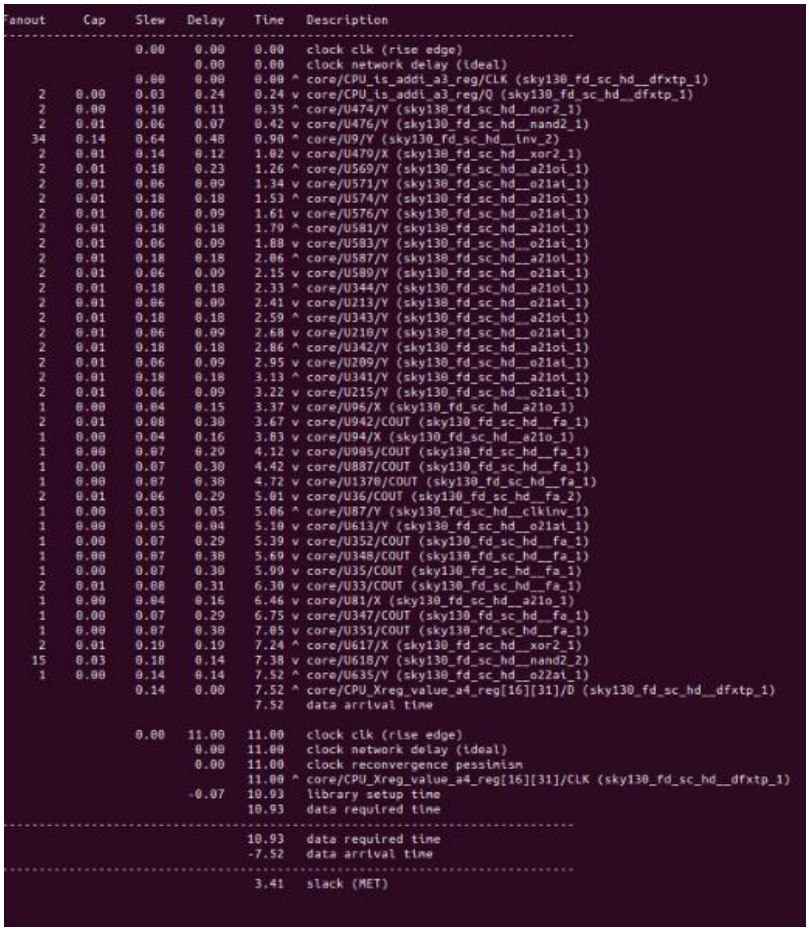


```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.03    0.27    0.27 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.12    0.13    0.39 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.06    0.08    0.48 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.14    0.71    0.54    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.15    0.15    1.16 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.20    0.27    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    2.35 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    2.65 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    2.96 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    3.06 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    3.26 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    3.36 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.20    0.21    3.57 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.10    3.67 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.04    0.17    3.84 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.09    0.35    4.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.18    4.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.33    4.71 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35    5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35    5.41 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.07    0.34    5.74 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.03    0.06    5.80 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.05    0.05    5.85 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.08    0.34    6.18 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35    6.53 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.35    6.88 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.36    7.24 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.18    7.42 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.08    0.33    7.76 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.34    8.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.22    0.22    8.32 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
    15    0.03    0.20    0.16    8.48 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
     1    0.00    0.15    0.16    8.64 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
                  0.15    0.00    8.64 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.64   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                         0.00   11.00   clock network delay (ideal)
                         0.00   11.00   clock reconvergence pessimism
                                11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
                        -0.08   10.92   library setup time
                                10.92   data required time
---------------------------------------------------------------------------
                                10.92   data required time
                                -8.64   data arrival time
---------------------------------------------------------------------------
                                 2.28   slack (MET)
```

12. sky130_fd_sc_hd__ff_n40C_1v65.lib

The library name identifies the Sky130 high-density standard-cell library at the Fast-Fast (`ff`) process corner, −40°C (`n40C`), and 1.65 V (`1v65`). The reported maximum-delay path is from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[16][31]`. Its data arrival time is 7.52 ns and the required time is 10.93 ns, giving a slack of +3.41 ns. OpenSTA marks the path as `MET`, so there is no setup violation. Compared with the 1.56 V FF corner, increasing the voltage reduces the arrival time by 1.12 ns and improves slack by 1.13 ns. This agrees with the expected effect of increased supply voltage on cell drive strength. The path remains the same endpoint as the 1.56 V case, but the individual gate delays are smaller, allowing the signal to arrive comfortably before the required time.

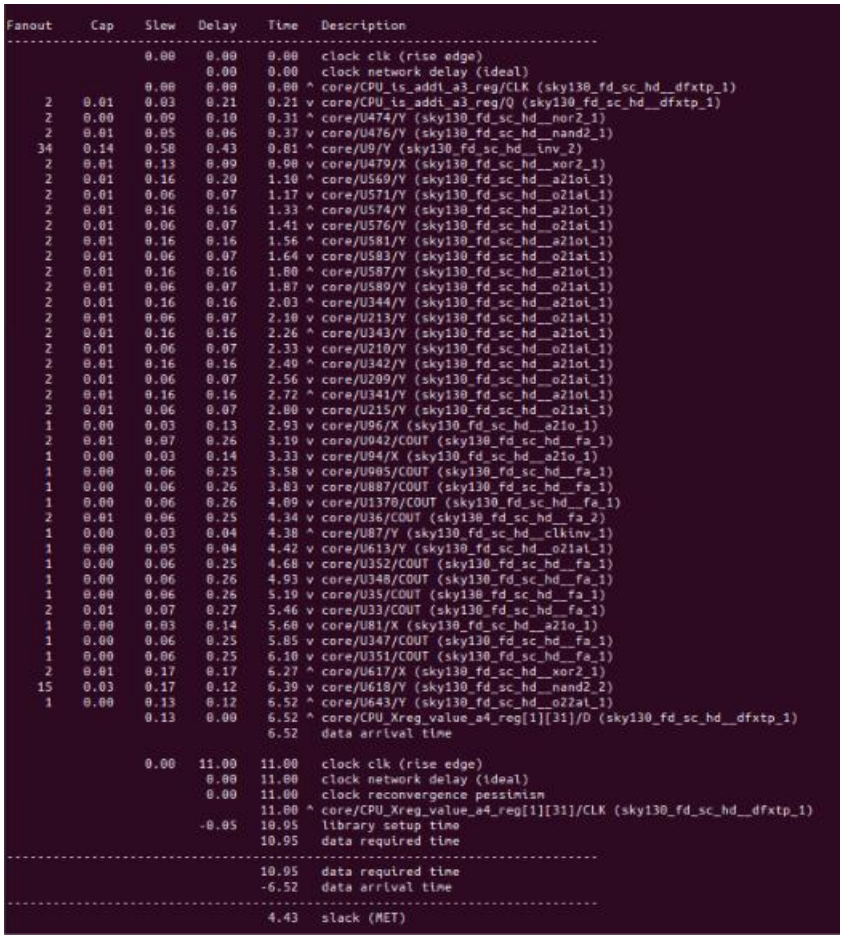


```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.03    0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.10    0.11    0.35 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.06    0.07    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.14    0.64    0.48    0.90 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.14    0.12    1.02 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.18    0.23    1.26 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    2.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    2.15 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    2.33 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    2.41 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    2.59 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    2.68 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    2.86 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    2.95 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.18    0.18    3.13 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.06    0.09    3.22 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.04    0.15    3.37 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.08    0.30    3.67 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.16    3.83 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.07    0.20    4.12 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.30    4.42 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.30    4.72 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.06    0.29    5.01 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.03    0.05    5.06 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.05    0.04    5.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.07    0.29    5.39 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.30    5.69 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.30    5.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.08    0.31    6.30 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.16    6.46 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.07    0.29    6.75 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.30    7.05 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.19    0.19    7.24 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
    15    0.03    0.18    0.14    7.38 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
     1    0.00    0.14    0.14    7.52 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
                0.14    0.00    7.52 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
                                7.52   data arrival time

                0.00   11.00   11.00   clock clk (rise edge)
                        0.00   11.00   clock network delay (ideal)
                        0.00   11.00   clock reconvergence pessimism
                               11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -0.07   10.93   library setup time
                               10.93   data required time
-----------------------------------------------------------
                               10.93   data required time
                               -7.52   data arrival time
-----------------------------------------------------------
                                3.41   slack (MET)
```

13. sky130_fd_sc_hd__ff_n40C_1v76.lib

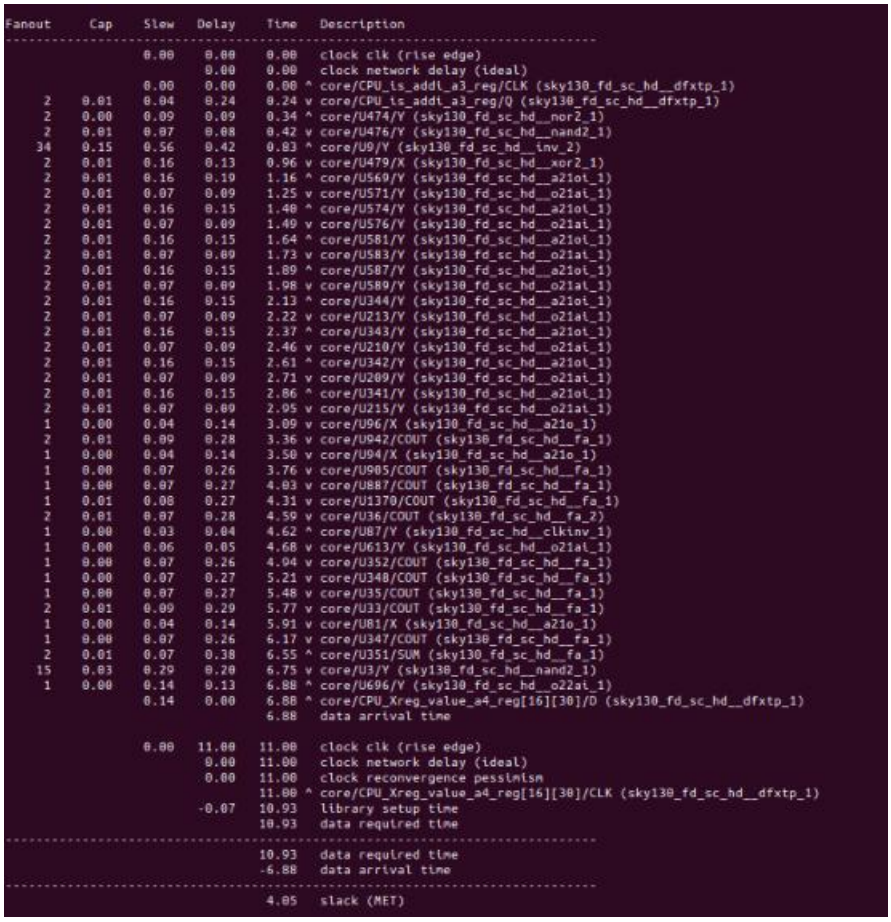
The library name represents the Sky130 high-density standard-cell library at the Fast-Fast (`ff`) process corner, −40°C (`n40C`), and 1.76 V (`1v76`). The maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[1][31]`. The data arrival time is 6.52 ns and the required time is 10.95 ns, resulting in a slack of +4.43 ns. The report therefore shows `slack (MET)` with no setup violation. Compared with the 1.65 V FF corner, the higher voltage reduces the arrival time by 1.00 ns and increases the available timing margin by 1.02 ns. The endpoint also changes, indicating that the most limiting path selected by OpenSTA is dependent on the corner-specific cell delays. Overall, this corner provides a substantially faster path than the SS corners, with more than four nanoseconds of setup margin on the reported path.



Fanout	Cap	Slew	Delay	Time	Description
		0.00	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
		0.00	0.00	0.00 ^	core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.03	0.21	0.21 v	core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.09	0.10	0.31 ^	core/U474/Y (sky130_fd_sc_hd__nor2_1)
2	0.01	0.05	0.06	0.37 v	core/U476/Y (sky130_fd_sc_hd__nand2_1)
34	0.14	0.58	0.43	0.81 ^	core/U9/Y (sky130_fd_sc_hd__inv_2)
2	0.01	0.13	0.09	0.90 v	core/U479/X (sky130_fd_sc_hd__xor2_1)
2	0.01	0.16	0.20	1.10 ^	core/U569/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	1.17 v	core/U571/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	1.33 ^	core/U574/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	1.41 v	core/U576/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	1.56 ^	core/U581/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	1.64 v	core/U583/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	1.80 ^	core/U587/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	1.87 v	core/U589/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	2.03 ^	core/U344/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	2.10 v	core/U213/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	2.26 ^	core/U343/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	2.33 v	core/U210/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	2.49 ^	core/U342/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	2.56 v	core/U209/Y (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.16	2.72 ^	core/U341/Y (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.06	0.07	2.80 v	core/U215/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00	0.03	0.13	2.93 v	core/U96/X (sky130_fd_sc_hd__a21o_1)
2	0.01	0.07	0.26	3.19 v	core/U042/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.03	0.14	3.33 v	core/U94/X (sky130_fd_sc_hd__a21o_1)
1	0.00	0.06	0.25	3.58 v	core/U905/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.06	0.26	3.83 v	core/U887/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.06	0.26	4.09 v	core/U1370/COUT (sky130_fd_sc_hd__fa_1)
2	0.01	0.06	0.25	4.34 v	core/U36/COUT (sky130_fd_sc_hd__fa_2)
1	0.00	0.03	0.04	4.38 ^	core/U87/Y (sky130_fd_sc_hd__clkinv_1)
1	0.00	0.05	0.04	4.42 v	core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00	0.06	0.25	4.68 v	core/U352/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.06	0.26	4.93 v	core/U348/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.06	0.26	5.19 v	core/U35/COUT (sky130_fd_sc_hd__fa_1)
2	0.01	0.07	0.27	5.46 v	core/U33/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.03	0.14	5.60 v	core/U81/X (sky130_fd_sc_hd__a21o_1)
1	0.00	0.06	0.25	5.85 v	core/U347/COUT (sky130_fd_sc_hd__fa_1)
1	0.00	0.06	0.25	6.10 v	core/U351/COUT (sky130_fd_sc_hd__fa_1)
2	0.01	0.17	0.17	6.27 ^	core/U617/X (sky130_fd_sc_hd__xor2_1)
15	0.03	0.17	0.12	6.39 v	core/U618/Y (sky130_fd_sc_hd__nand2_2)
1	0.00	0.13	0.12	6.52 ^	core/U643/Y (sky130_fd_sc_hd__o22ai_1)
		0.13	0.00	6.52 ^	core/CPU_Xreg_value_a4_reg[1][31]/D (sky130_fd_sc_hd__dfxtp_1)
				6.52	data arrival time
		0.00	11.00	11.00	clock clk (rise edge)
			0.00	11.00	clock network delay (ideal)
			0.00	11.00	clock reconvergence pessimism
				11.00 ^	core/CPU_Xreg_value_a4_reg[1][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
			-0.05	10.95	library setup time
				10.95	data required time
				10.95	data required time
				-6.52	data arrival time
				4.43	slack (MET)

14. sky130_fd_sc_hd__ff_100C_1v65.lib

The library name denotes the Sky130 high-density standard-cell library at the Fast-Fast (`ff`) process corner, 100°C (`100C`), and 1.65 V (`1v65`). The reported maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[16][30]`. The data arrival time is 6.88 ns and the data required time is 10.93 ns, giving a slack of +4.05 ns. The path therefore meets the setup requirement with 4.05 ns of positive margin and has no reported violation. Compared with the −40°C FF 1.65 V result, the endpoint changes and the arrival time is 0.64 ns smaller in this particular report. This demonstrates that the actual characterized timing can vary with temperature and that the same nominal voltage and process label does not guarantee identical path behavior. The important result is that the reported path remains comfortably inside the timing limit.

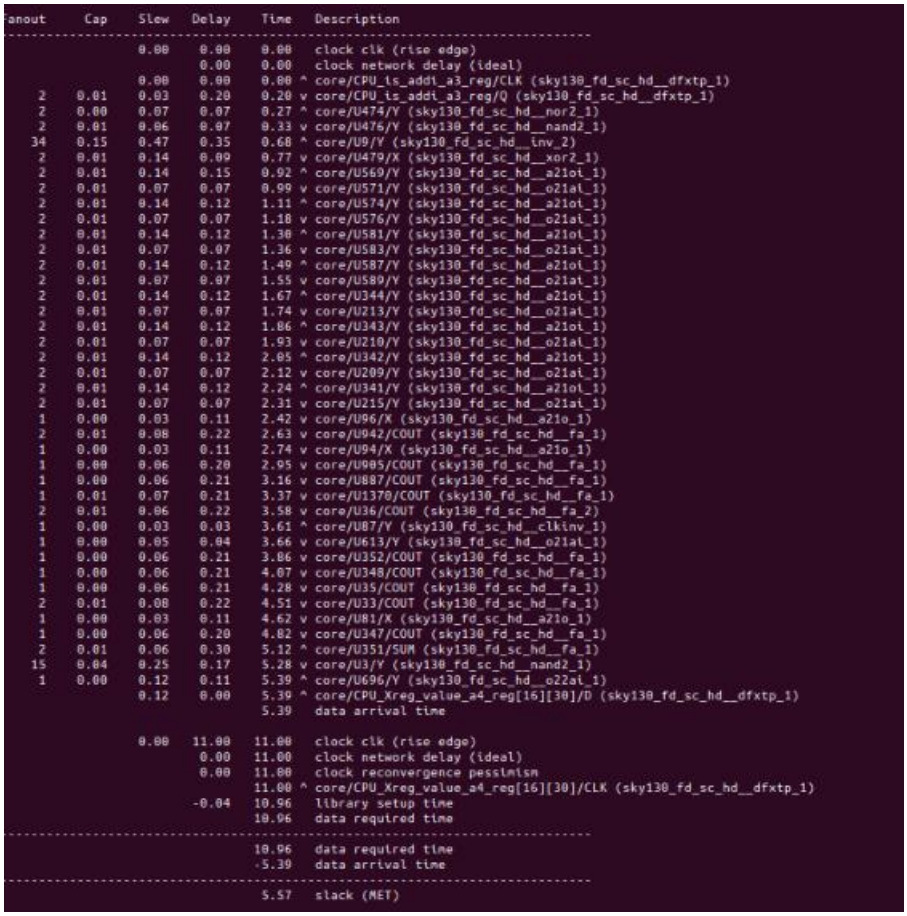


```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.09    0.09    0.34 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
     2    0.01    0.07    0.08    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    34    0.15    0.56    0.42    0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
     2    0.01    0.16    0.13    0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.16    0.19    1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    1.89 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    1.98 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    2.13 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    2.22 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    2.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    2.46 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    2.61 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    2.71 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.15    2.86 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.07    0.09    2.95 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.04    0.14    3.09 v core/U96/X (sky130_fd_sc_hd__a21o_1)
     2    0.01    0.09    0.28    3.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.14    3.50 v core/U94/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.07    0.26    3.76 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.27    4.03 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
     1    0.01    0.08    0.27    4.31 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.07    0.28    4.59 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
     1    0.00    0.03    0.04    4.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.06    0.05    4.68 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.07    0.26    4.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.27    5.21 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.07    0.27    5.48 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.29    5.77 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
     1    0.00    0.04    0.14    5.91 v core/U81/X (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.07    0.26    6.17 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
     2    0.01    0.07    0.38    6.55 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
    15    0.03    0.29    0.20    6.75 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.14    0.13    6.88 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
                  0.14    0.00    6.88 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          6.88    6.88   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.07   10.93   library setup time
                                 10.93   data required time
-----------------------------------------------------------------------------
                                 10.93   data required time
                                 -6.88   data arrival time
-----------------------------------------------------------------------------
                                  4.05   slack (MET)
```

15. sky130_fd_sc_hd__ff_100C_1v95.lib

The library name identifies the Sky130 high-density standard-cell library at the Fast-Fast (`ff`) process corner, 100°C (`100C`), and 1.95 V (`1v95`). This is the highest-voltage FF corner in the supplied set. The reported maximum-delay path runs from `core/CPU_is_addi_a3_reg` to `core/CPU_Xreg_value_a4_reg[16][30]`. Its data arrival time is 5.39 ns and the required time is 10.96 ns, resulting in a slack of +5.57 ns. OpenSTA reports `slack (MET)`, so there is no setup violation for this path. The high supply voltage and FF process result in very small cell delays, giving the largest positive setup margin among the reported FF corners. Compared with the 1.65 V FF corner at 100°C, increasing the voltage to 1.95 V reduces the arrival time by 1.49 ns and improves slack by 1.52 ns. Thus the actual output clearly demonstrates the strong timing improvement associated with the higher-voltage FF characterization.



```
fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_ls_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
    2    0.01     0.03    0.20    0.20 v core/CPU_ls_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
    2    0.00     0.07    0.07    0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
    2    0.01     0.06    0.07    0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   34    0.15     0.47    0.35    0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
    2    0.01     0.14    0.09    0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
    2    0.01     0.14    0.15    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    1.67 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    1.86 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    2.05 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
    2    0.01     0.14    0.12    2.24 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
    2    0.01     0.07    0.07    2.31 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
    1    0.00     0.03    0.11    2.42 v core/U96/X (sky130_fd_sc_hd__a21o_1)
    2    0.01     0.08    0.22    2.63 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
    1    0.00     0.03    0.11    2.74 v core/U94/X (sky130_fd_sc_hd__a21o_1)
    1    0.00     0.06    0.20    2.95 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
    1    0.00     0.06    0.21    3.16 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
    1    0.01     0.07    0.21    3.37 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
    2    0.01     0.06    0.22    3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
    1    0.00     0.03    0.03    3.61 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
    1    0.00     0.05    0.04    3.66 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
    1    0.00     0.06    0.21    3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    1    0.00     0.06    0.21    4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    1    0.00     0.06    0.21    4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    2    0.01     0.08    0.22    4.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    1    0.00     0.03    0.11    4.62 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    1    0.00     0.06    0.20    4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    2    0.01     0.06    0.30    5.12 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
   15    0.04     0.25    0.17    5.28 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
    1    0.00     0.12    0.11    5.39 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
                  0.12    0.00    5.39 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
                          5.39    data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                         0.00   11.00   clock network delay (ideal)
                         0.00   11.00   clock reconvergence pessimism
                                11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                        -0.04   10.96   library setup time
                                10.96   data required time
-----------------------------------------------------------------------------
                                10.96   data required time
                                -5.39   data arrival time
-----------------------------------------------------------------------------
                                 5.57   slack (MET)
```

SUMMARY OF REPORTED SLACK

==========

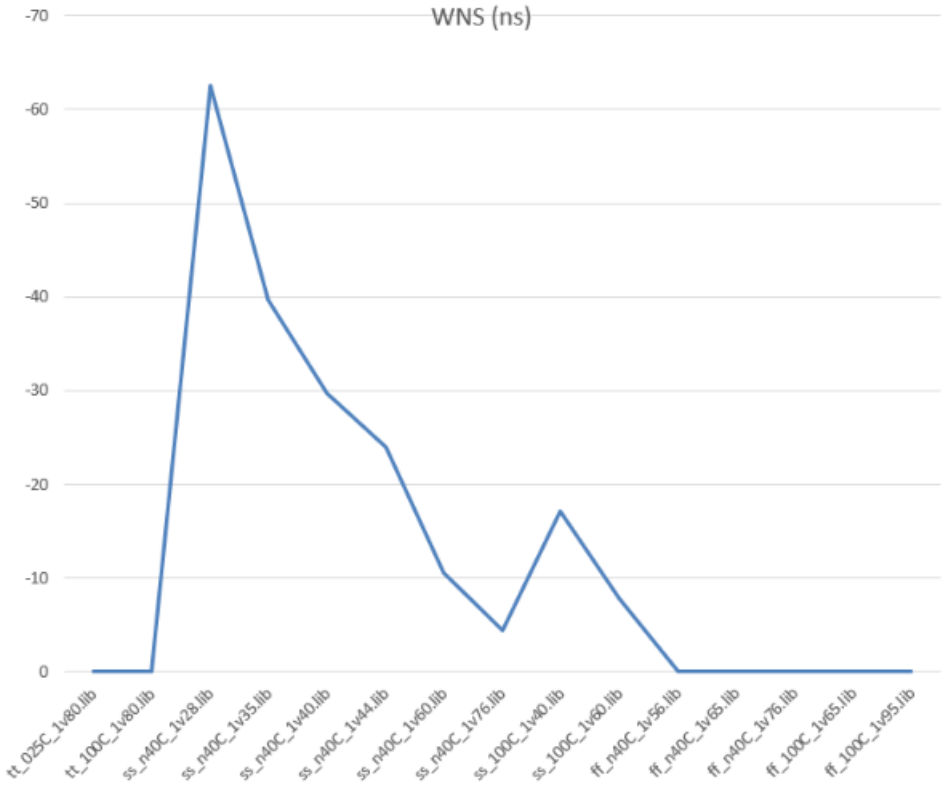
TT corners: +1.94 ns and +2.09 ns (both MET).

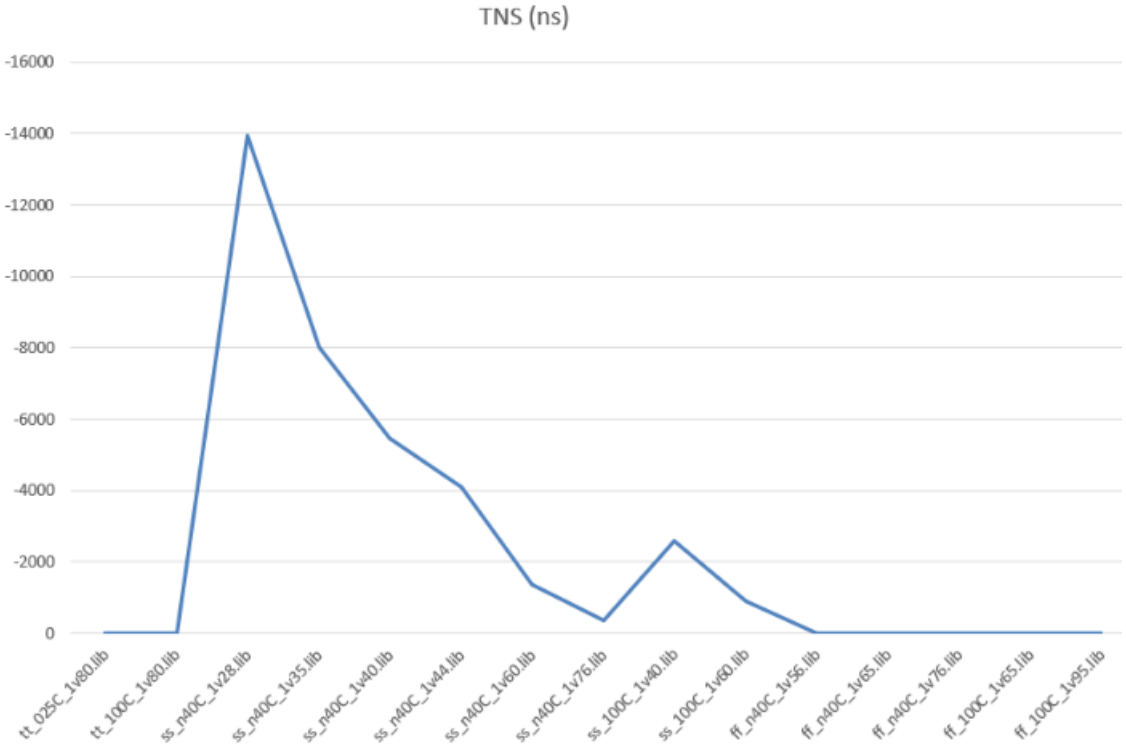
SS corners: -62.56, -39.70, -29.69, -23.91, -10.61, -4.38, -17.12, and -7.87 ns (all VIOLATED).

FF corners: +2.28, +3.41, +4.43, +4.05, and +5.57 ns (all MET).

The supplied results therefore show large setup violations in every reported SS corner, while the TT and FF reported paths have positive setup slack. The SS -40°C results improve steadily as voltage rises from 1.28 V to 1.76 V, and the FF results generally show substantially smaller path delays and positive margin.

WNS and TNS


WNS (ns)
-70
-60
-50
-40
-30
-20
-10
0
tt_025C_1v80.lib
tt_100C_1v80.lib
ss_n40C_1v28.lib
ss_n40C_1v35.lib
ss_n40C_1v40.lib
ss_n40C_1v44.lib
ss_n40C_1v60.lib
ss_n40C_1v76.lib
ss_100C_1v40.lib
ss_100C_1v60.lib
ff_n40C_1v56.lib
ff_n40C_1v65.lib
ff_n40C_1v76.lib
ff_100C_1v65.lib
ff_100C_1v95.lib


TNS (ns)
-16000
-14000
-12000
-10000
-8000
-6000
-4000
-2000
0
tt_025C_1v80.lib
tt_100C_1v80.lib
ss_n40C_1v28.lib
ss_n40C_1v35.lib
ss_n40C_1v40.lib
ss_n40C_1v44.lib
ss_n40C_1v60.lib
ss_n40C_1v76.lib
ss_100C_1v40.lib
ss_100C_1v60.lib
ff_n40C_1v56.lib
ff_n40C_1v65.lib
ff_n40C_1v76.lib
ff_100C_1v65.lib
ff_100C_1v95.lib

Library	Process	Temperature	Voltage	WNS (ns	TNS (ns)	Status
tt_025C_1v80.lib	TT	25°C	1.80 V	0	0	MET
tt_100C_1v80.lib	TT	100°C	1.80 V	0	0	MET
ss_n40C_1v28.lib	SS	-40°C	1.28 V	-62.56	-13957.59	VIOLATED
ss_n40C_1v35.lib	SS	-40°C	1.35 V	-39.7	-7993.79	VIOLATED
ss_n40C_1v40.lib	SS	-40°C	1.40 V	-29.69	-5466.48	VIOLATED
ss_n40C_1v44.lib	SS	-40°C	1.44 V	-23.91	-4107.19	VIOLATED
ss_n40C_1v60.lib	SS	-40°C	1.60 V	-10.61	-1355.8	VIOLATED
ss_n40C_1v76.lib	SS	-40°C	1.76 V	-4.38	-367.93	VIOLATED
ss_100C_1v40.lib	SS	100°C	1.40 V	-17.12	-2599.51	VIOLATED
ss_100C_1v60.lib	SS	100°C	1.60 V	-7.87	-892.7	VIOLATED
ff_n40C_1v56.lib	FF	-40°C	1.56 V	0	0	MET
ff_n40C_1v65.lib	FF	-40°C	1.65 V	0	0	MET
ff_n40C_1v76.lib	FF	-40°C	1.76 V	0	0	MET
ff_100C_1v65.lib	FF	100°C	1.65 V	0	0	MET
ff_100C_1v95.lib	FF	100°C	1.95 V	0	0	MET

WNS – Worst Negative Slack

WNS (Worst Negative Slack) is a timing metric used in Static Timing Analysis (STA) to identify the **single timing path with the largest timing failure**.

In a digital circuit, a signal must reach its destination within a specified time. STA calculates the **slack** of each path by comparing when the signal actually arrives with when it is required to arrive.

$$Slack = Required\ Time - Actual\ Arrival\ Time$$

For setup analysis:

- **Positive slack** means the signal arrives before the required time.
- **Zero slack** means the signal arrives exactly at the required time.
- **Negative slack** means the signal arrives later than the required time.

WNS is obtained by finding the **smallest slack value** among all the analyzed paths:

$$WNS = Minimum\ Slack$$

Example

Suppose five timing paths have the following slack values:

Path	Slack
Path 1	+5 ns
Path 2	-2 ns
Path 3	-8 ns
Path 4	-3 ns
Path 5	+1 ns

The WNS is:

$$WNS = -8\ ns$$

This means that **Path 3 has the largest individual timing violation**, missing its required timing by 8 ns.

Therefore, WNS is mainly useful for answering:

“What is the worst timing problem in the design?”

A WNS value closer to zero indicates that the worst path has a smaller timing violation.

A negative WNS indicates that at least one path does not satisfy the timing requirement.

TNS – Total Negative Slack

TNS (Total Negative Slack) gives a broader view of timing problems by considering **all paths that have negative slack** rather than only looking at the worst path.
To calculate TNS, the negative slack values of all violating paths are added together:

$$TNS = \sum_{i=1}^{N} N\,egative\,Slack_i$$

For example, assume the circuit has these timing paths:

Path	Slack
Path 1	+4 ns
Path 2	-2 ns
Path 3	-5 ns
Path 4	-3 ns
Path 5	+1 ns

Only the negative values are considered:

$$TNS = (-2) + (-5) + (-3)$$
$$TNS = -10\,ns$$

So, the TNS is **-10 ns**.
The positive-slack paths do not contribute to TNS because they already satisfy the timing requirement.
TNS therefore answers a different question:
"How much timing violation exists when all failing paths are considered together?"

Difference Between WNS and TNS
Although both metrics are related to negative timing slack, they describe **different aspects of the timing problem**.

Feature	WNS	TNS
Full form	Worst Negative Slack	Total Negative Slack
What it considers	Worst individual path	All failing paths
Purpose	Finds the most severe single violation	Measures the overall amount of violation
Calculation	Smallest slack value	Sum of all negative slack values
Example	-8 ns	-10 ns

Consider these slack values:

$$-2,\ -5,\ -3$$

The **WNS is -5 ns**, because -5 ns is the worst individual value.
The **TNS is -10 ns**, because:

$$-2 - 5 - 3 = -10\,ns$$

So, **WNS focuses on one path, while TNS considers the complete set of failing paths**.

An important point

A very negative TNS does **not necessarily mean that one path has an extremely large delay**.

For example:

- 100 paths each having -2 ns slack

would give:

$$TNS = 100 \times (-2) = -200\,ns$$

while:

$$WNS = -2\,ns$$

This shows that the design may have **many relatively small timing violations**, even though its worst individual violation is only -2 ns.

Conversely, a design could have:

$$WNS = -20\,ns$$

but only one failing path, giving:

$$TNS = -20\,ns$$

Thus, looking at both values gives a more complete picture.

What WNS and TNS Tell Us in STA

During STA, these metrics help identify whether the design satisfies its timing constraints.

If WNS = 0 ns

There is no negative slack in the analyzed paths, so no setup timing violation is being reported.

If WNS < 0 ns

At least one path has failed its timing requirement. The further WNS moves below zero, the larger the violation on that particular worst path.

If TNS = 0 ns

There are no paths contributing negative slack.

If TNS < 0 ns

Multiple paths may be violating timing, or one or more paths may have significant violations. The value represents their **combined negative slack**.

run.tcl:

```
set libs {
  sky130_fd_sc_hd__tt_025C_1v80.lib
  sky130_fd_sc_hd__tt_100C_1v80.lib

  sky130_fd_sc_hd__ss_n40C_1v28.lib
  sky130_fd_sc_hd__ss_n40C_1v35.lib
```

```
    sky130_fd_sc_hd__ss_n40C_1v40.lib
    sky130_fd_sc_hd__ss_n40C_1v44.lib
    sky130_fd_sc_hd__ss_n40C_1v60.lib
    sky130_fd_sc_hd__ss_n40C_1v76.lib

    sky130_fd_sc_hd__ss_100C_1v40.lib
    sky130_fd_sc_hd__ss_100C_1v60.lib

    sky130_fd_sc_hd__ff_n40C_1v56.lib
    sky130_fd_sc_hd__ff_n40C_1v65.lib
    sky130_fd_sc_hd__ff_n40C_1v76.lib

    sky130_fd_sc_hd__ff_100C_1v65.lib
    sky130_fd_sc_hd__ff_100C_1v95.lib
}

foreach lib $libs {

    puts "\n========================================="
    puts "LIBRARY: $lib"
    puts "========================================="

    read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/$lib
    read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/$lib

    read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/avsddac.lib
    read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/avsddac.lib

    read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdpll.lib
    read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdpll.lib

    read_verilog /data/Desktop/OpenSTA/vsdbabysoc/vsdbabysoc.synth.v

    link_design vsdbabysoc

    read_sdc /data/Desktop/OpenSTA/vsdbabysoc/sdc/vsdbabysoc_synthesis.sdc

    report_checks //report_wns and report_tns for wns and tns
}
```